Data Availability Statement

For the manuscript: "What a radical-pair quantum reservoir would require: readable capacity, no quantum advantage, and a nuclear-register criterion"

Hikaru Wakaura and Taiki Tanimae, QIRI (Quantum Integrated Research Institute Inc.), Tokyo, Japan

`h.wakaura@qiri.co.jp`, `t.tanimae@qiri.co.jp`

Code. All results in this work are produced by the open-source Python package `qbscreen` (MIT licence), available at https://github.com/deeptell-inc/new_Q_brain. The package implements the exact Liouville-space Haberkorn master equation with electronic decoherence and the correlated-pair-birth quantum-reservoir-computing protocol. The specific modules are:

- `qbscreen/master_equation.py` — Liouville-space solver and its validation against the closed-form coherent limit (ESI, S1);
- `qbscreen/reservoir.py` — reservoir Hamiltonian and propagator, and the out-of-sample information-processing capacity (IPC) and memory capacity (MC) estimators (trained on the first half of each run, scored on the held-out second half);
- `qbscreen/corrected_injection.py` — the chemically consistent correlated-pair birth channel $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$ and its S–T_0 coherent variant, together with the single-electron-reset control that reproduces the artifact it replaces;
- `qbscreen/readout_routes.py` — the seven biologically accessible readout routes with recombination active, the product-weighted CIDNP observable, the turnover clock, and the nuclear-register depolarisation parameter q;
- `qbscreen/final_numbers.py` — protocol-consistent recomputation of every headline quantity reported in the main text and ESI;
- `qbscreen/qrc_benchmarks.py` — classical echo-state-network baselines, the NARMA2 benchmark, and the coherence-time scan;
- `manuscript/make_schematic.py` and `manuscript/make_fig_criterion.py` — Figs. 1–3.

Data. All numerical outputs underlying the figures and tables are provided as JSON files in `simulation_results/` of the repository, principally `final_numbers.json`, `corrected_injection.json`, `readout_routes.json` `register_reuse.json`, `grid_convergence.json`, `qrc_benchmarks.json` and `cryptochrome_reali` No experimental data were generated; the study is entirely computational. Every reported capacity is out of sample, is quoted with its shuffled-input null floor, and is shown to be converged in sample length; values are means over 4–10 input realisations with the stated standard deviations; the turnover-clock scan uses a single realisation, since the quantity of interest there is the ratio of memory capacities across pause lengths on the same input.

Reproducibility. The central claims—the agreement of the Liouville-space solver with the coherent limit, the Dambre capacity bound, the correlated-pair-birth readout capacities, and the nuclear-register reuse criterion—are covered by an automated test suite (`qbscreen/tests/`; 28 tests, all passing). All figures and numbers can be regenerated from a clean checkout with

```
pip install -e .
python -m pytest qbscreen/tests/ -q
python -m qbscreen.final_numbers
python -m qbscreen.readout_routes
```

Software. Python 3, NumPy (≥ 1.24), SciPy (≥ 1.10) and Matplotlib (≥ 3.7). No proprietary software is required.

This statement follows the Royal Society of Chemistry Data Sharing Policy: `https://www.rsc.org/journals-books-databases/about-journals/data-sharing-policy/`